

CHAYMA ROUAA

MECHANICAL DESIGN | multidisciplinary environment welcomed

Seeking an apprenticeship | September 2026 | One-year contract

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PROFILE

Mechanical engineering student at Université Paris-Saclay, with a focus on mechanical design, robotics, mechatronics and numerical simulation. Experience in CAD design, robotic mechanisms, kinematics, trajectory control and scientific modelling. Proficient in SolidWorks, CATIA, MATLAB, Python and Fortran, with a strong foundation in solid mechanics, dynamics, vibrations and stability.

EDUCATION



B. Sc Year 3 - Engineering Sciences

Université Evry Paris-Saclay

Key Coursework

Sensors and Actuators, Automation, Signal Processing, Design and Manufacturing, Digital Control and Control Systems, Operating Systems



B. Sc - Mechanical Engineering

Sorbonne Université, Paris

Key Coursework

Mechanical Design, Electronics, Strength of Materials, Fluid Mechanics, Thermodynamics, Solid Mechanics, Continuum Mechanics, Equilibrium and Stability, Robotic Systems, Prototyping

PROJECTS

Plantigrade Robot - Chebyshev Linkage

SolidWorks CAD design of a plantigrade quadruped robot based on a Chebyshev four-bar linkage. Parametric mechanical modelling, assemblies, motion study, ground and stair simulation, motor actuation, gravity, mass properties, cost analysis and technical drawings.

3R Robotic Arm - MATLAB

Robotic modelling and trajectory control of a three-revolute-joint arm. Forward and inverse kinematics, Jacobian matrix, joint-space and Cartesian interpolation, trajectory planning, continuous trajectory tracking and velocity control.

Crack Propagation Simulation - Python

Computational modelling of crack propagation in a crystalline material. Particle generation, atomic configurations, neighbour detection with matscipy and Lennard-Jones interactions, followed by scientific trajectory visualisation.

Granular Flow Modeling - Fortran 90

Particle-based numerical modelling of granular flow under gravity, including particle generation, neighbour interactions, temporal evolution and trajectory visualisation.

Liquid Damper - Vibrational Dynamics

Python simulation of mechanical oscillations and dynamic response for vibration mitigation in large structures subjected to external disturbances.

Snap-Through Stability Mechanism - Mathematica

Mechanical analysis of elastic structures with multiple equilibrium states using equilibrium and stability theory.

Adjustable Mirror Support - CATIA

Multi-degree-of-freedom mechanical design considering stability, mechanical architecture and manufacturability.

Motor-Assisted Transport Cart

Mechanical and mechatronic system design integrating stabilisation, motor assistance, dynamic start-up propulsion, directional control and emergency motor stop.

Parametric Hairpin - SolidWorks

Parametric two-piece CAD design with curved-surface modelling and wrapping on curved surfaces, with iterative resolution of geometric and manufacturing constraints.

Interactive Lily - TouchDesigner | MediaPipe

Parametric modelling of flower growth using Bézier curves and the Verhulst logistic model, combined with real-time human-machine interaction using hand tracking with MediaPipe.

TECHNICAL SKILLS

Mechanical Design & CAD

SolidWorks · CATIA · Parametric 3D Modelling · Mechanical Assemblies · 2D Technical Drawings · ISO GPS · Tolerance Stack-Up · Section Views · Exploded Views · Motion Studies · Mass Properties · DfAM · Manufacturability · Prototyping

Numerical Simulation & Programming

Python · MATLAB · Fortran 90 · Numerical Modelling · Computational Mechanics · Particle-Based Simulation · Atomistic Modelling · Neighbour Detection · Lennard-Jones Interactions · Data Analysis · Scientific Visualisation

Robotics & Mechatronics

Robotic Mechanisms · Kinematics · Forward Kinematics · Inverse Kinematics · Jacobian Matrix · Trajectory Planning · Joint-Space Interpolation · Cartesian Interpolation · Trajectory Tracking · Velocity Control · Dynamic Systems · Mechanical Actuation

Mechanical Engineering

Solid Mechanics · Strength of Materials · Continuum Mechanics · Structural Mechanics · Dynamics · Vibrations · Equilibrium · Stability · Fluid Mechanics · Thermodynamics · Requirements Specification · Mechanical Analysis · Experimental Analysis · Problem Solving · Technical Writing